

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458642.92726 +/- 0.00092 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.036
Transit depth (Rp/Rs)^2	2.4e-05 +/- 0.0035 [%]
Orbital Inclination (inc)	87.11 +/- 1.78
Airmass coefficient 1 (a1)	1.201 +/- 0.062
Airmass coefficient 2 (a2)	-0.17 +/- 0.2
Scatter in the residuals of the lightcurve fit is	5.15 %
Best Comparison Star	None
Optimal Aperture	3.7
Optimal Annulus	4.81
Transit Duration (day)	0.0824 +/- 0.005

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.0 ± 0.036 vs 0.13031 (deviation 2.5σ)
Duration fit vs published	0.0824 d vs 0.0954 d (deviation 1.8σ)
Mid-time O-C	-1.4 min
Depth SNR	0.0
Residual scatter	5.15 %

Light curve

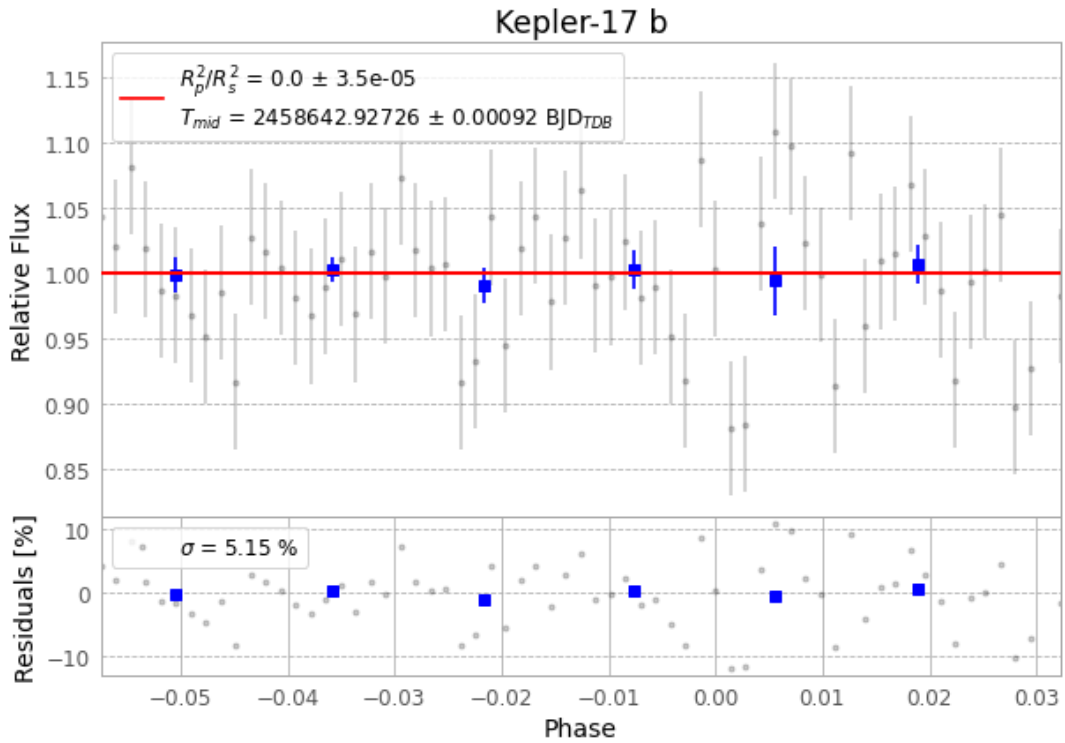


Figure 1 — FinalLightCurve_Kepler-17 b_2019-06-08.png (SHA-256 76cdf2a31cedec7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [417, 169], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1dc2cf0d49971823...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

64 FITS frames (42 MB), KEPLER-17190608080912.FITS → KEPLER-17190608112115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-KEPLER-17-190608.log (05ded5304c52...), FinalLightCurve_Kepler-17 b_2019-06-08.png (76cdf2a31ce...), FinalLightCurve_Kepler-17 b_2019-06-08.csv (6b49b108276c...)

Compute resources

Wall time 113.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-19 01:06Z.